

math111a hw2

jc — October 09, 2025

Problem 1

Let a and b be commuting elements of a group G . Prove that $(ab)^n = a^n b^n \forall n \in \mathbb{Z}$

Solution:

We can recall the multiplicative $(ab)^n$ notation from class where $n \in \mathbb{Z}$

Case 1: $n > 0$

Consider that for $n = 1$ we have that $(ab)^1 = ab$ and that $a^1 b^1 = ab$

Assume that $(ab)^n = a^n b^n$ holds for some $n > 0$ then we would like to show that $(ab)^{n+1} = a^{n+1} b^{n+1}$

Note that this is equivalent to showing that $(ab)^n ab = a^n ab^n b$ and by assumption, $(ab)^n ab = a^n b^n ab$

Since a and b commute, we have that $ab^n = b^n a$ and the claim holds by induction

Case 2: $n = 0$ then we have $(ab)^0 = 1$ and $a^0 b^0 = (1)(1) = 1$ by identity property for groups

Case 3: $n < 0$

First recall that $(ab)^{-1} = b^{-1} a^{-1}$ and that $a^n a^{-1} = \underbrace{a \cdot \dots \cdot a}_{n \text{ times}} a^{-1} = \underbrace{a \cdot \dots \cdot a}_{n-1 \text{ times}} \cdot e = a^{n-1}$

Next note that $ab = ba \Rightarrow (ab)^{-1} = (ba)^{-1} \Rightarrow b^{-1} a^{-1} = a^{-1} b^{-1}$ (i)

Therefore for $n = -1$ we have $(ab)^{-1} = b^{-1} a^{-1} = a^{-1} b^{-1}$

Assume that $(ab)^n = a^n b^n$ holds for some $n < 0$ then we would like to show that $(ab)^{n-1} = a^{n-1} b^{n-1}$

Note that $(ab)^{n-1} = (ab)^n (ab)^{-1} = a^n b^n (ab)^{-1} = a^n b^n b^{-1} a^{-1} = a^n b^n a^{-1} b^{-1}$

and that $a^{n-1} b^{n-1} = a^n a^{-1} b^n b^{-1}$

To show that $b^n a^{-1} = a^{-1} b^n$ consider that $ab = ba \Rightarrow aba^{-1} = b \Rightarrow ba^{-1} = a^{-1} b$

Combining these facts, the claim holds by induction

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(i) Since the inverse is unique, we can be sure that if $a = b$ then $a^{-1} = b^{-1}$

Problem 2

Let G be a group. Prove that if $a^2 = 1$ for all $x \in G$, then G is Abelian

Corrected version: Let G be a group. Prove that if $x^2 = 1$ for all $x \in G$, then G is Abelian

Solution:

Let $a, b \in G$

Note that $a^2 = aa = 1 \Rightarrow a = a^{-1}$ (every element is its own inverse)

Then we have:

$$ab = a^{-1}b^{-1} = (ba)^{-1} = ba$$

where the second equality was proven in class. ■

Problem 3

Prove that in any group, an element and its inverse have the same order

Solution:

Let $|a|$ denote the order of a and note that $(a^{-1})^n = (a^n)^{-1}$ and note that the inverse of identity is itself

Let $a \in G$ with order n then $a^n = 1 \Rightarrow (a^n)^{-1} = (1)^{-1} \Rightarrow (a^{-1})^n = 1 \Rightarrow |a^{-1}| \leq |a|$

Similarly if n is now the inverse of a^{-1} then $(a^{-1})^n = 1 \Rightarrow (a^{(-1)^n})^{-1} = (1)^{-1} \Rightarrow a^n = 1 \Rightarrow |a| \leq |a^{-1}|$

This means that if a has finite order, then $|a| = |a^{-1}|$

If a has infinite order then $\nexists k \in \mathbb{Z}, k > 0$ such that $a^k = 1$ then from the above we know that some $k > 0$ does not exist for $(a^{-1})^k = 1$ either.

Another way to say this is that if a has infinite order and assume that a^{-1} has finite order. In the finite case we showed that if a^{-1} has finite order then a has the same (finite) order, which is a contradiction and therefore a^{-1} must also have infinite order.

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references: https://proofwiki.org/wiki/Powers_of_Group_Elements/Negative_Index

Problem 4

In class we studied that $D_{2n} = \langle r, s \mid r^n = 1, s^2 = 1, sr = r^{-1}s \rangle$. Prove that for all $k \in \mathbb{Z}$, $sr^k = r^{-k}s$

Solution:

Case 1: $k = 0$ then $sr^0 = r^{-0}s \Rightarrow s = s$

Case 2: $k \geq 1$

For $k = 1$ we have that $sr = r^{-1}s$ which holds by the presentation of D_{2n}

Assuming that $sr^k = r^{-k}s$ holds for some $k \geq 1$ then

$$\begin{aligned} sr^{k+1} &= r^{-(k+1)}s \\ sr^k r &= r^{-k} r^{-1} s \\ r^{-k} sr &= r^{-k} r^{-1} s && \text{by assumption} \\ sr &= r^{-1} s && \text{by definition} \end{aligned}$$

Case 3: $k \leq -1$

For $k = -1$ we have $sr^{-1} = rs$ which holds since $sr = r^{-1}s \Rightarrow rsr = s \Rightarrow rs = sr^{-1}$

Assuming that $sr^k = r^{-k}s$ holds for some $k \leq 1$ then

$$\begin{aligned} sr^{k-1} &= r^{-(k-1)}s = r^{-k+1}s \\ sr^k r^{-1} &= r^{-k} rs \\ r^{-k} sr^{-1} &= r^{-k} rs && \text{by assumption} \\ sr^{-1} &= rs && \text{as proved earlier} \end{aligned}$$

Then through induction on all possible k the claim follows

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Problem 5

In the group D_{2n} explain why

- a) a reflection followed by another reflection must be a rotation
- b) a reflection and a rotation taken together in either order must be a reflection

Solution:

Note that a reflection in the group D_{2n} is of the form sr^a where $a \in \{0, \dots, n-1\}$ and a rotation is of the form r^b where $b \in \{0, \dots, n-1\}$

Let $k_1, k_2 \in \{0, \dots, n-1\}$ then:

- a) $sr^{k_1}sr^{k_2} = r^{-k_1}ssr^{k_2} = r^{k_2-k_1} \in D_{2n}$ and
 - b) $r^{k_1}sr^{k_2} = r^{k_1}r^{-k_2}s = r^{k_1-k_2}s = sr^{k_2-k_1} \in D_{2n}$ and
- $$sr^{k_2}r^{k_1} = sr^{k_2+k_1} \in D_{2n}$$

We can justify the three inclusions since we can reduce the powers of r mod n

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<https://math.stackexchange.com/questions/4837202/arithmatic-on-elements-of-d-2n#:~:text=You%20have%20the%20right%20idea,Add%20a%20comment>

Problem 6

Write out Cayley tables for the groups S_3 and D_8

Solution:

S_3 :

$\circ$	$()$	$(1\ 2)$	$(1\ 3)$	$(2\ 3)$	$(1\ 2\ 3)$	$(1\ 3\ 2)$
$()$	$()$	$(1\ 2)$	$(1\ 3)$	$(2\ 3)$	$(1\ 2\ 3)$	$(1\ 3\ 2)$
$(1\ 2)$	$(1\ 2)$	$()$	$(1\ 2\ 3)$	$(1\ 3\ 2)$	$(1\ 3)$	$(2\ 3)$
$(1\ 3)$	$(1\ 3)$	$(1\ 3\ 2)$	$()$	$(1\ 2\ 3)$	$(2\ 3)$	$(1\ 2)$
$(2\ 3)$	$(2\ 3)$	$(1\ 2\ 3)$	$(1\ 3\ 2)$	$()$	$(1\ 2)$	$(1\ 3)$
$(1\ 2\ 3)$	$(1\ 2\ 3)$	$(2\ 3)$	$(1\ 2)$	$(1\ 3)$	$(1\ 3\ 2)$	$()$
$(1\ 3\ 2)$	$(1\ 3\ 2)$	$(1\ 3)$	$(2\ 3)$	$(1\ 2)$	$()$	$(1\ 2\ 3)$

$(1\ 2)$

D_8 :

handwritten

Problem 7

Prove that for all $n \geq 3$, the group S_n is not Abelian

Solution:

Let $\sigma, \tau \in S_3$ and let $\sigma = (12), \tau = (23)$

Computing their compositions we obtain $\sigma\tau = (123)$ and $\tau\sigma = (132)$ so that S_3 is not abelian

Next consider the groups S_n where $n \geq 4$

We can construct $\sigma^\sim, \tau^\sim \in S_{n \geq 4}$ such that they act exactly like σ, τ on the elements $\{1, 2, 3\}$ and add additional mappings such that $\sigma^\sim(i) = i, \tau^\sim(i) = i$ for $i \in \{4, \dots, n_{\geq 4}\}$

Therefore $\sigma^\sim\tau^\sim$ and $\tau^\sim\sigma^\sim$ will act the same as in S_3 on elements $\{1, 2, 3\}$ and not commute under $\circ$ either.

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Problem 8

Let σ be the permutation

$$1 \mapsto 3$$

$$2 \mapsto 4$$

$$3 \mapsto 5$$

$$4 \mapsto 2$$

$$5 \mapsto 1$$

and let τ be the permutation

$$1 \mapsto 5$$

$$2 \mapsto 3$$

$$3 \mapsto 2$$

$$4 \mapsto 4$$

$$5 \mapsto 1$$

Find the cycle decompositions of each of the following permutations: $\sigma, \tau, \sigma^2, \sigma\tau, \tau\sigma, \tau^2\sigma$

Solution:

$$\sigma: (135)(24)$$

$$\tau: (15)(23)(4)$$

$$\sigma^2: (153)(2)(4)$$

$$\tau\sigma: (1243)(5)$$

$$\sigma\tau: (1)(2534)$$

$$\tau^2\sigma: (135)(24) = \sigma$$